

PROJECT EXECUTION & MONITORING

From Spud-in to Rig Release: Mastering Operational Control

Safety Moment

Before we start

- Know the nearest emergency exit
- Assembly point outside the building
- First aider & AED location
- Phones on silent

Today's topic: Line of Fire

Many rig floor injuries happen when a person is in the path of moving or released energy.

- Suspended loads & pipe handling
- Pressurised lines during testing
- Rotating equipment & tongs
- Stored energy: springs, chains, slings

Ask the class: where were you last in the line of fire, and what moved you out of it?

SAFETY MOMENT

TOPIK HARI INI: LINE OF FIRE !

Banyak cedera di lantai rig terjadi saat seseorang berada di jalur pergerakan atau pelepasan energi.



Beban Tergantung & Penanganan Pipa

Hindari berada di bawah atau di jalur ayunan pipa/beban.



Jalur Bertekanan Saat Pengujian

Jauhi jalur pipa/selang saat testing atau saat tekanan tinggi.



Peralatan Berputar & Tongs

Waspadai bagian mesin yang berputar dan arah sapuan kunci tong.



Energi Terakumulasi: Pegas, Rantai, Sling

Jauhi zona lenting (*snapback*) jika sling atau rantai putus tiba-tiba.

JAGA KOMUNIKASI DAN KEKOMPAKAN, KESEHATAN-KESELAMATAN-DAN LINGKUNGAN KERJA #1

KESELAMATAN ADALAH TANGGUNG JAWAB KITA BERSAMA.
BEKERJA AMAN, PULANG SELAMAT!

Agenda

1	**Opening**	Safety moment & Review Phase-2	**10'**
2	**Phase-3**	Execution & Monitoring	**45'**
3	**Q&A + Break**		**20'**
4	**Phase-4**	Project Closing	**30'**
5	**Assignment**	Drilling Program & DWOP format	**15'**
6	**Group Work**	Draft Drilling Program with mentor	**45'**
7	**Closing**	Mandate & DWOP schedule	**15'**

180'
total

Review DPM Phase-2

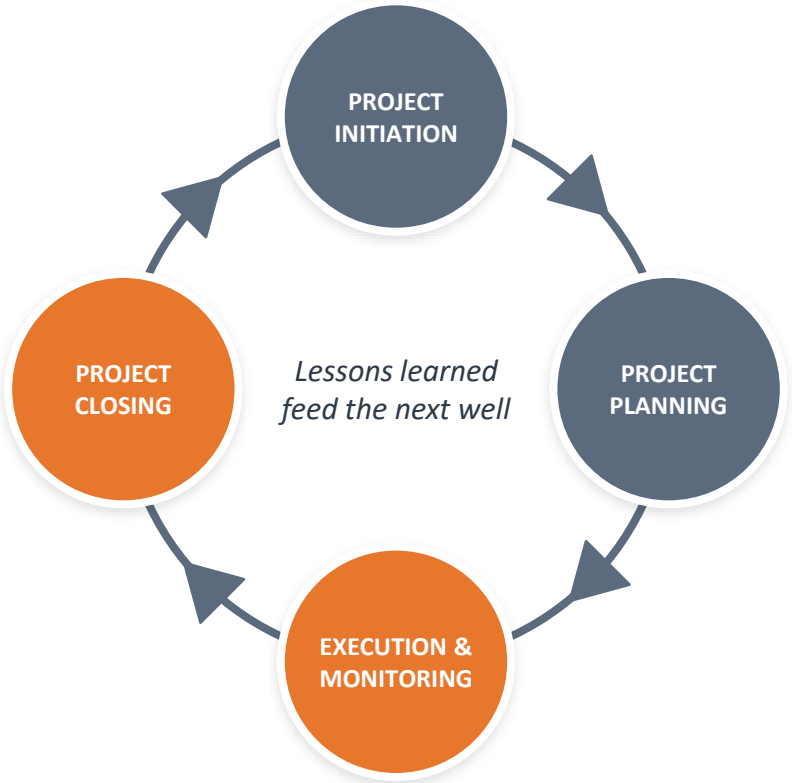
Catatan Peer Review	Fokus untuk Phase-3
1 BOD vs Drilling Program Beberapa kelompok masih bingung membedakan BOD dan Drilling Program.	Pahami perbedaannya (slide berikut): BOD = apa & mengapa, Program = bagaimana, langkah demi langkah.
2 Konsistensi data Masih banyak data yang tidak konsisten antar bagian (PPFG, casing depth, mud weight, hari, biaya).	Gunakan satu data sheet bersama untuk semua role dan lakukan cross-check sebelum submit.
3 Penguasaan materi Banyak yang hafal slide tetapi belum menguasai isinya, sehingga bingung saat ditanya.	Setiap anggota harus bisa menjelaskan alasannya. Di DWOP, panel bertanya langsung per role.
4 Kelengkapan dokumen BOD seharusnya sampai Rig Release, tetapi banyak yang berhenti di pemasangan production casing.	BOD wajib mencakup seluruh sumur sampai completion/suspend, rig down, dan Rig Release. Drilling Program mengikuti scope yang sama.

BOD vs Drilling Program

	Basis of Design (BOD)	Drilling Program
Question answered	WHAT will we build and WHY	HOW will we drill it, step by step
Phase / Gate	Phase-2 Planning → Peer Review	Phase-3 Execution → DWOP
Main audience	Engineers, peer reviewers, management	Rig team: Company Man, Toolpusher, service crews
Level of detail	Design basis, calculations, options chosen	Instructions, parameters, procedures per section
Typical content	PPFG, casing design, trajectory, risk register, AFE	Operational sequence, mud & cement program, BOP tests, MOC
Scope	Location preparations to P&A	Spud to rig release (full well)
Changes after approval	Re-design & re-review	Through MOC

Rule of thumb: the BOD convinces the reviewer the design is right; the Drilling Program tells the rig crew exactly what to do.

Where We Are in the Project Life Cycle



1. **Initiation**
   Drilling Proposal → Gate 1 Approval
2. **Planning**
   Basis of Design → Peer Review
3. **Execution & Monitoring**
   Drilling Program, DWOP, DDR, MOC
4. **Closing**
   End of Well Report, Lessons Learned

Today's focus: the PM shifts from designer to controller, then closes the project so its lessons make the next well better.

Strategic Execution

The Core of Real-Time Operational Excellence

The Execution Core

PLAN

Approved Drilling Program, DWOP outcomes, P50 Days-vs-Depth & AFE baseline

EXECUTE

Rig operations per the step-by-step sequence, with clear roles at the wellsite

MONITOR

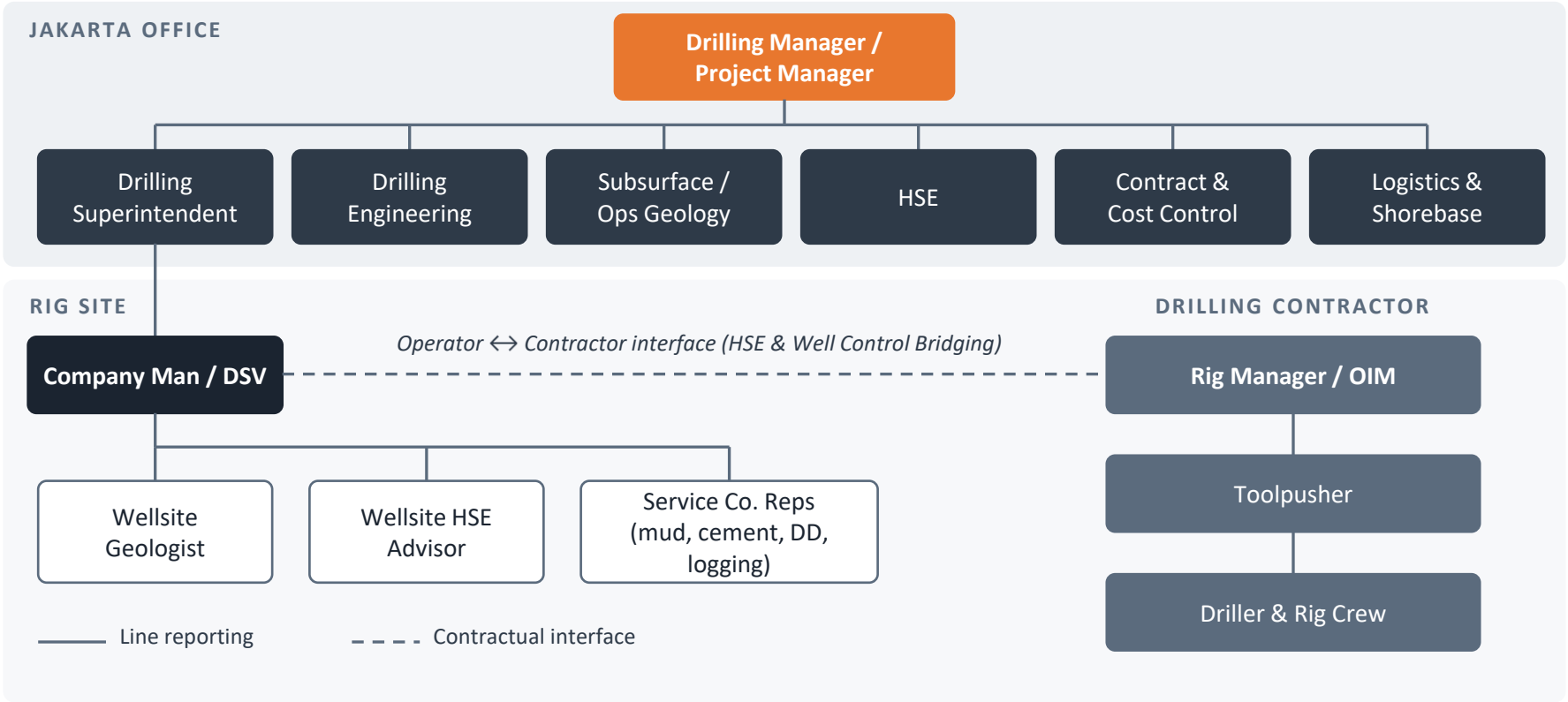
DDR, real-time data and KPIs compared against the baseline every 24 hours

CONTROL

Corrective action, MOC and escalation when actuals drift from plan

Key message: execution is a daily closed loop. The baseline set in Phase-2 is the yardstick; every deviation must be seen early, explained, and acted upon.

Execution Organization Structure



Business Partners (Serco) on Location

Service Line	Key Personnel	Rig-up	Surface	Interm.	Prod.	Logging	Compl. & Test
Drilling Contractor	Rig Manager, Toolpusher, crew	■	■	■	■	■	■
Drilling Fluids	Mud Engineer		■	■	■	■	
Solids Control & Waste	SCE / Waste Operator		■	■	■	■	■
Mud Logging	Mud Logger, Data Engineer		■	■	■	■	
Directional & MWD/LWD	DD, MWD Engineer			■	■		
Casing Running	TRS Crew		▒	▒	▒		▒
Cementing	Cementing Engineer & Crew		▒	▒	▒		
Wellhead	Wellhead Technician		▒	▒	▒		▒
Wireline Logging	Logging Engineer			▒		■	
Completion & Well Test	Completion / DST Crew						■
Support Services	Catering, medic, security, transport	■	■	■	■	■	■

■ On location full-time ▒ Call-out per job

Every serco on site: contract/call-off active, HSE bridging, induction, daily report & signed field ticket.

Execution Readiness: Documents Before Spud

Technical

Drilling Program
approved & signed

DWOP report
actions closed out

Service programs
mud, cement, directional, logging

Contingency plans
losses, stuck pipe, shallow gas, H2S

HSE

HSE Bridging Document
Operator ↔ Drilling Contractor

Well Control Bridging
shut-in & kill responsibilities

ERP
MEDEVAC, oil spill, H2S response

Rig acceptance report
inspection findings closed

Regulatory & Permits

Approved AFE
& work program (SKK Migas)

Drilling permit
notification to regulator

Environmental approval
AMDAL / UKL-UPL

Land & certification
land use, valid equipment certificates

Commercial & Logistics

Contracts & call-offs
rig and all service lines active

BOM & LLI tracker
material on location

Rig move plan
route, permits, schedule

DDR & cost tracker
templates and contact list

Regulatory Permits for Drilling (Indonesia)

Before Spud

Persetujuan Lingkungan
AMDAL / UKL-UPL (PP 22/2021)

PPKH
if location is in a forest area (KLHK)

Land & community
land acquisition, local approvals

AFE & WP&B approval
SKK Migas

Installation & equipment safety
PLO / inspection (Permen ESDM 32/2021)

During Drilling

Izin Handak
purchase, use, storage, transport: Polri, with SKK Migas purchase approval

Explosives magazine
Ditjen Migas

Radioactive source licence
well logging: BAPETEN

Limbah B3
temporary storage & handling approvals

Heavy-load transport
rig move route permits

Closing

Unused explosives
return, transfer or destruction permit

Radioactive sources
return & transport records

Environmental reporting
RKL-RPL & site restoration

Final well reports
SKK Migas & Ditjen Migas

Well status notification
suspension or P&A

Critical path: explosives and radioactive permits have long lead times; start early and verify current regulations.

Road to Spud: Readiness Timeline

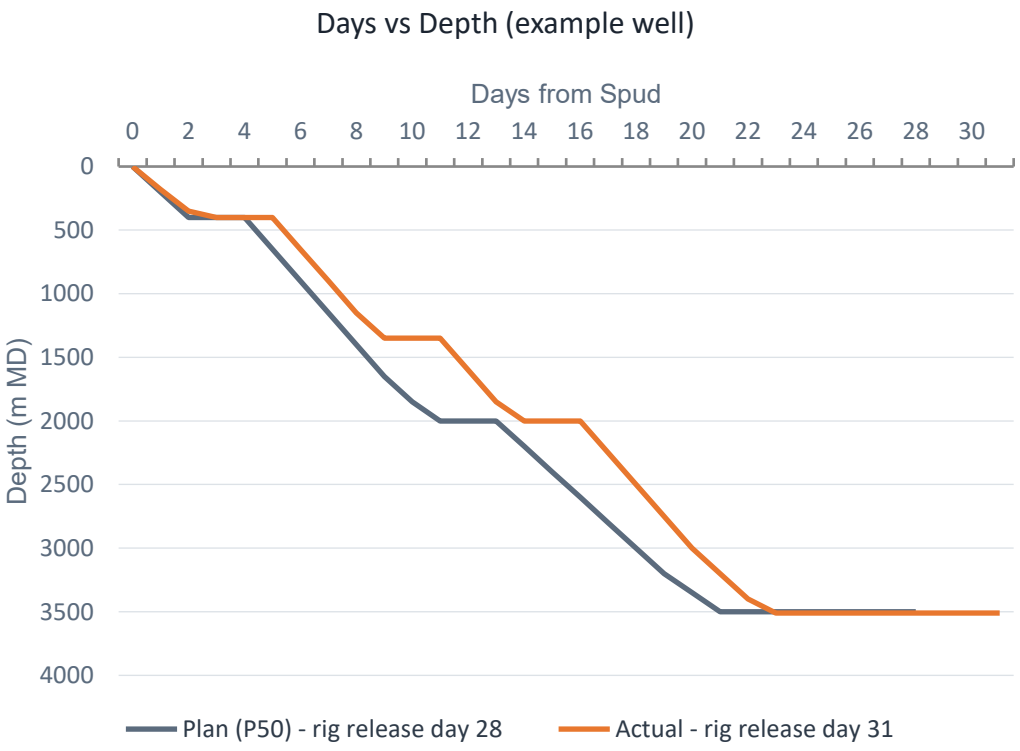


Pre-spud Go / No-Go checks

- All permits valid and on location
- Every document approved and signed
- Crew briefed; well control drill done
- Material for first sections on site

No document, no spud.
Missing approvals become NPT on day one, at full spread rate.

Performance Tracking



What the PM watches daily

- **Days vs Depth:** actual against P50 (and P10/P90 band)
- **ROP:** on-bottom drilling speed per section
- **Flat time:** casing, cementing, BOP, logging
- **Tripping & connections:** the hidden ILT source
- **Unit cost:** US$/m and days/1,000 m

Example: TD reached day 23 vs plan day 21 — two days lost to lost circulation in the 12-¼" section.

Non-Productive Time (NPT)

Productive Time

Operations that move the well toward its objective as planned

NPT

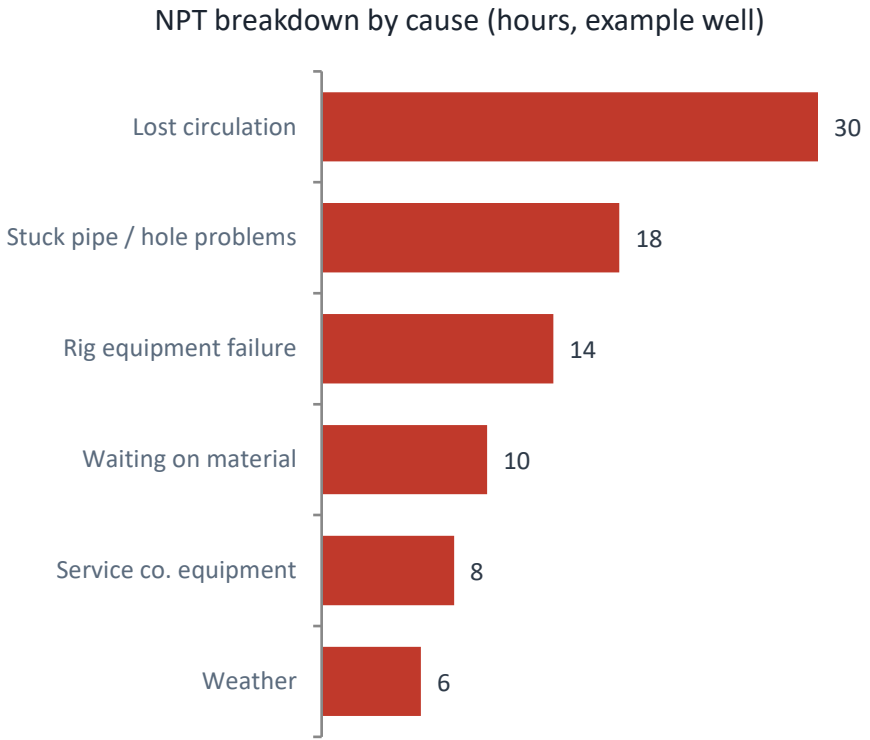
Time lost to unplanned events: failures, hole problems, waiting

ILT

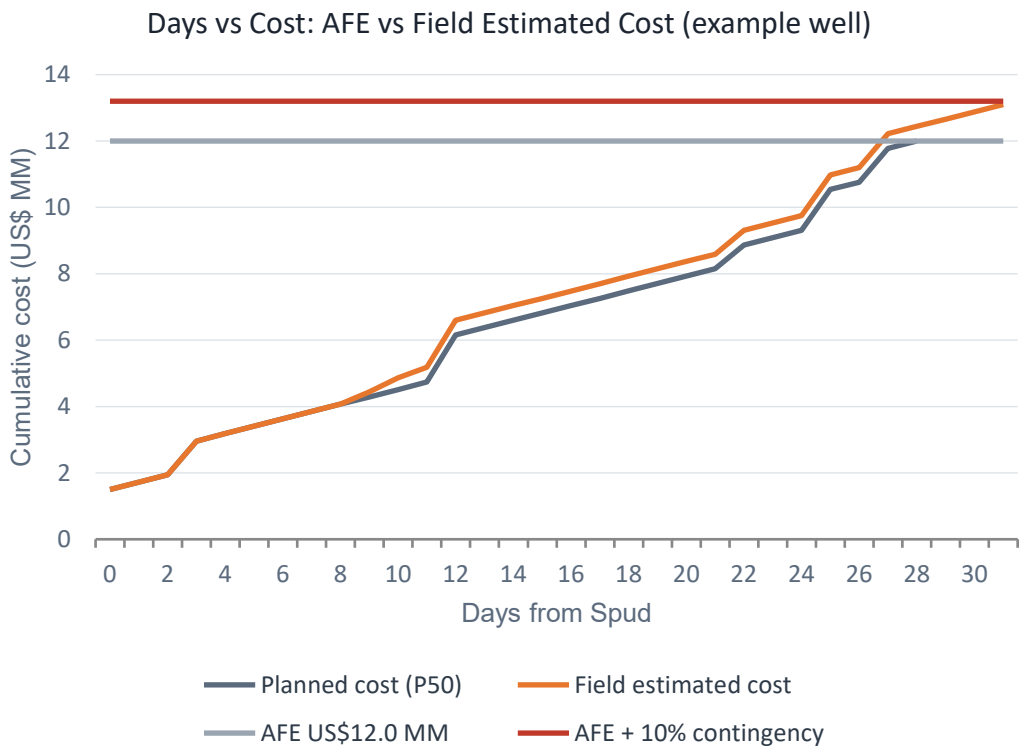
Invisible Lost Time: planned work done slower than best practice

86 hrs ≈ 3.6 days NPT

≈ 11.6% of well time × spread rate = direct cost



Cost Control: AFE Monitoring



Daily Cost

Rig spread + services + consumables

Variance

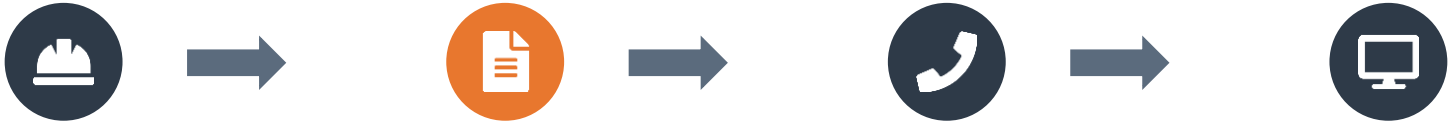
Actual cost-to-date – Plan cost-to-date

EAC

Cost-to-date + Estimate-to-complete

- EAC > AFE: early warning to management
- EAC > AFE + contingency: supplementary AFE before spending

Information Flow



Wellsite

Company Man, Driller, Mud Logger, Service Crews

DDR @ 06:00

Operations, time breakdown, NPT, daily cost

Morning Call

Ops, Engineering, Subsurface, Logistics align on next 24 hrs

Management

Dashboard, weekly report, partners & regulator

Real-Time Dashboard

The Daily Drilling Report (DDR) is the primary source of truth. Leveraging real-time data allows PMs to make proactive decisions instead of reacting to past failures.

- Morning Calls (Operations Update)
- 24/7 Digital Monitoring
- AFE vs. Field Estimated Cost

Wellsite Reporting Requirements

Company Man / DSV (Operator)

Report	Frequency
Daily Drilling Report (DDR)	Daily, 06:00
Daily cost / field estimated cost	Daily
POB & material inventory	Daily
Incident / near-miss flash report	Immediately
MOC request	As required
BOP test, casing & cement summary	Per event
Weekly HSE report	Weekly
End of section report	Per section

Business Partner Engineers (Service Cos.)

Report	Frequency
IADC daily drilling report (Toolpusher)	Daily
Daily mud report (Mud Engineer)	Daily
Mud log & geological report	Daily
Directional survey report (DD/MWD)	Per survey / daily
Cementing job report	Per job
Bit & BHA run report	Per run
Wireline / logging report	Per run
Field / service ticket	Per job

Flow: partner reports feed the DDR; signed field tickets are the basis for invoice reconciliation.

Coordination Meeting Cadence

Frequency	Meeting	Where	Focus
Daily	**Morning Call**	Office ↔ Rig	Last 24 hrs, next 24 hrs, issues
Weekly	**Weekly Safety Meeting**	Rig site	HSE performance, HOC review, drills
Weekly / Bi-weekly	**Business Partner Coordination Meeting**	Jakarta office	Progress, cost, HSE, look-ahead, issues
Monthly	**Management Review**	Jakarta office	KPIs, AFE status, key decisions

Attendees: operator PMT, drilling contractor, key service companies, and partners as required.



Partner Meeting Agenda

- HSE performance & HOC trends
- Progress vs P50 plan
- Cost vs AFE & invoicing
- 2-week look-ahead & logistics
- Service quality & open issues
- MOC & decisions needed

Regulatory Reporting During Operations

Report	To	When
Spud notification	SKK Migas, Ditjen Migas	At spud
Daily operations report	SKK Migas	Daily
Progress & AFE realization	SKK Migas	Monthly
Incident / accident notification	Ditjen Migas (Inspection), SKK Migas	Immediately, then full report
Environmental monitoring (RKL-RPL)	Environmental authority	Periodic (per approval)
Well status change: suspend / P&A	SKK Migas, Ditjen Migas	Before execution



Who prepares

- Company Man: operations data
- Cost Controller: AFE realization
- HSE: incident & environment
- PM: reviews & signs off

Formats and deadlines follow current SKK Migas PTK and Ditjen Migas rules; always check the latest version.

Look-Ahead Plan (7-Day Example)

Day	Planned Operation	Depth (m)	Materials & Services	Risks / Notes	Status
D+1	Drill 12-¼" with LCM on standby	1,600 – 1,850	LCM pills, mud, DD/MWD	Losses zone	Ready
D+2	Drill to section TD; circulate & POOH	2,000	Mud logging, DD	Tight hole on trip	Ready
D+3	Run 9-⅝" casing	2,000	Casing crew, 9-⅝" casing on site	Running speed, surge	Ready
D+4	Cement 9-⅝" casing; WOC	2,000	Cement unit, bulk cement	Losses during cement job	Pending
D+5	N/U & test BOP; drill out; FIT	2,000	BOP test unit, 8-½" BHA	Well control barrier test	Ready
D+6	Drill 8-½" section	2,000 – 2,250	Barite stock, DD/MWD	Pressure ramp, kick tolerance	Pending
D+7	Drill 8-½" section	2,250 – 2,500	Bit #6 on location	Bit wear	Pending

7-day: updated daily, reviewed in the morning call

14-day: updated weekly for logistics & partner meeting

Logistic Plan: Shorebase & Transportation



Shorebase Plan

- **Warehouse & yard:** casing, wellhead, LLI storage and inspection
- **Bulk & fluids:** mud plant, cement silos, fuel and water
- **Jetty / staging area:** load-out, manifest, weighing
- **Shorebase team:** material control & expediting 24/7

Transportation Plan

- **Land:** route & bridge survey, heavy-haul permits, convoy & journey management
- **Marine:** supply vessels, crew boats, weather windows
- **Air:** helicopter crew change & MEDEVAC standby
- **Personnel:** crew change schedule, POB control

Social Issues: Onshore vs Offshore

Onshore

Land & compensation
disputes over land, crops, access roads

Blockades & demonstrations
road or location closed by locals

Local workforce & suppliers
demand for local hire and local vendors

Nuisance
dust, noise, heavy traffic, water sources

Adat & cultural sites
customary land, sacred places

Offshore

Fishermen (nelayan)
loss of fishing ground, exclusion zone

Damage claims
nets, fish traps (bagan, rumpon) hit by vessels

Marine traffic
conflict with local boats and shipping lanes

Spill concern
coastal communities worry about pollution

Shorebase community
local vessels, jobs and supplies at port

Common Controls

- Stakeholder mapping before mobilization
- Socialization with Pemda & community leaders
- Community Relations officer on site
- Grievance mechanism, fast response
- Local hire & local procurement plan
- PPM / CSR program aligned with SKK Migas
- Log social NPT in the DDR

Management of Change

Maintaining Stability in a Dynamic Environment

MOC Framework

1 **Request**

2 **Risk Assessment**

3 **Technical Review**

4 **Approval**

5 **Implement & Communicate**

6 **Close-out & Document**

Typical MOC triggers

- Casing point or TD change
- Mud weight outside the programmed window
- Trajectory change or sidetrack
- Equipment, vendor or material substitution
- Deviation from the approved operational sequence

Rules of the game

- **Authority:** approval level scales with risk and cost impact
- **Urgent MOC:** verbal approval by the PM, documented within 24 hrs
- **Never:** change first, ask later
- **Close-out:** update program, AFE and risk register

Key Decisions During Execution

Decision	Trigger / Data	Decided by	Plan B if not met
Spud	Permits, documents, rig acceptance	PM (pre-spud meeting)	Delay spud, close gaps
Casing point	Formation tops, pore pressure indicators, losses	Drilling Supt + Subsurface	Set casing early / contingency string
Well control response	Kick, losses, gas readings	Company Man on site	Kill, cure losses, escalate
TD call	Target reached, shows, mud log	Subsurface + PM	Drill deeper via MOC / stop short
Logging & coring	Shows, hole condition	Subsurface	Reduced suite / LWD only
Well status	Log & test results vs success criteria	Management	Complete / suspend / P&A

Link to Phase-1: these are the branches of the PM's decision tree. Define triggers and authority before spud, not at 02:00 on the rig floor.

HSE Program at the Rig

Meeting & communication rhythm

Pre-tour Safety Meeting

Every shift: hazards, plan of the day, lessons from previous tour

Pre-job Meeting & JSA

Before every critical or non-routine job

Weekly Safety Meeting

All crews, incident review, HSE KPI update

Safety Induction

Every new arrival on location, visitors included

Drills & observation

- Kick / BOP drills per crew
- H2S, fire & abandon-rig drills
- MEDEVAC & ERP exercise

- HOC (Hazard Observation Card)
- Stop Work Authority for everyone
- Daily HSE inspection & weekly audit

Permit to Work (PTW)

Request	Hazard ID & JSA	Isolation & Gas Test	Authorize	Execute & Monitor	Close-out & Handback

Work that requires a permit

Hot work
welding, cutting, grinding

Confined space
tanks, pits, cellar

Work at height
derrick, monkey board

Electrical / LOTO
isolation & lock-out

Critical lifting
heavy or blind lifts

Radioactive & explosives
logging sources, perforating

Key rules

- One permit, one job, one shift
- Toolbox talk before work starts
- SIMOPS checked against other permits
- Area handed back safe

SIMOPS: Simultaneous Operations

SIMOPS: two or more activities at the same location and time whose hazards can interact.

Typical in drilling

- Drilling next to producing wells on a cluster pad
- Rig move or skid near live wellheads
- Perforating with explosives near hot work or radios
- Crane lifts over live equipment
- Construction or hook-up beside the rig

Controls

SIMOPS plan & matrix · one person in charge · PTW cross-check · interface meeting

Example SIMOPS matrix

	Drill	Offline	Hot work	Perf.	Test
Drilling / tripping		OK	C	X	X
Offline BHA make-up	OK		C	X	C
Hot work	C	C		X	X
Perforating (explosives)	X	X	X		C
Well test / flaring	X	C	X	C	

OK Allowed · C With controls / PTW · X Not permitted

Matrix values are illustrative; each well has its own SIMOPS plan.

Well Control & Emergency Escalation

TIER 1

Rig Site

Driller shuts in; Company Man & Toolpusher run the kill plan

TIER 2

Jakarta Ops Center

Drilling Supt & PM support; well control specialist on call

TIER 3

Crisis Management

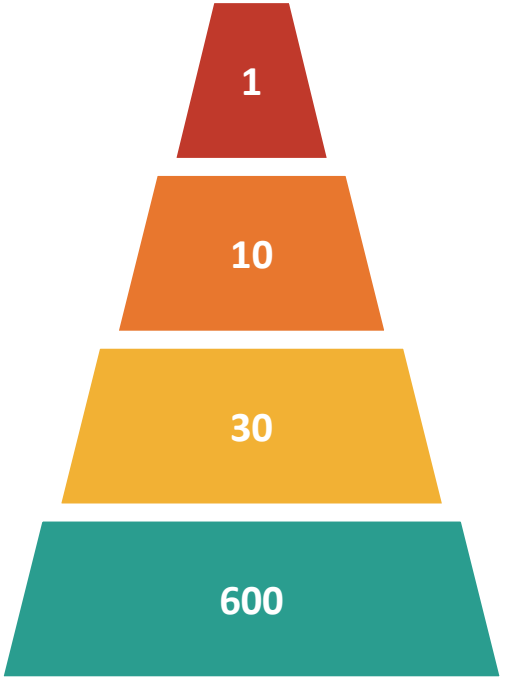
Management & external well control company; regulator & partners notified

Escalate when the event exceeds the capability of the current tier

First 15 minutes: kick

- Detect: pit gain, flow increase
- Shut in the well per bridging procedure
- Record SIDPP, SICP and pit gain
- Muster non-essential personnel
- Company Man calls Drilling Supt
- Prepare kill sheet & kill mud
- Log times for the incident report

Safety Performance



Major injury

Serious injuries

Minor injuries

Near misses

Bird's safety pyramid: shrink the base to prevent the top

Leading indicators

- Near-miss & HOC reports
- Stop Work Authority used
- Kick & H2S drills on schedule
- BOP tests on time
- Safety meetings & JSA

Lagging indicators

- Lost Time Injury (LTI)
- TRIR
- Well control incidents
- Spills & environmental events

TRIR = recordable cases × 200,000 ÷ man-hours

“No plan survives first contact with the subsurface. The role of the PM is to adapt the plan without losing the goal.”

- Adapted for DPM Excellence

Q&A

???



PROJECT CLOSING

Ensuring Accountability, Knowledge Capture, and Asset Handover

The Final Handover

Tying up loose ends for a sustainable legacy

Operational Closure

Final Well Status

Completed producer, suspended, or P&A with barrier diagram

Well Handover

Wellhead, tree & documents handed to Production with handover certificate

Rig Release & Demob

Rig and service equipment demobilized safely

Material Backload

Surplus casing, mud & cement returned or transferred

Site Restoration

Location cleaned, waste disposed, permits closed

Exit criterion: the asset owner signs the handover; responsibility for the well formally moves out of the drilling project.

Administrative & Financial

Financial Close-out

- **Invoice reconciliation:** every service ticket matched to DDR & contract
- **Accruals:** costs incurred but not yet invoiced
- **Final cost vs AFE:** variance explained line by line
- **AFE closure:** close the cost center, release unused budget

Administrative Close-out

- **Contract close-out:** settle claims, release guarantees
- **Vendor evaluation:** performance scorecards per service line
- **Material reconciliation:** inventory transferred or written off
- **Regulatory reporting:** final reports to regulator & partners

Business Partner Close-out Reports

Business Partner	Close-out Report	Feeds Into
Drilling Contractor	End of well summary: IADC time & rig downtime	EOWR, SQM
Drilling Fluids	End of well fluids report: volumes, losses, chemicals used, cost	EOWR, cost close-out
Cementing	Post-job reports per casing, job evaluation	EOWR, well integrity file
Directional & MWD/LWD	Definitive survey & final directional report	EOWR, regulator, anti-collision
Mud Logging / Geology	Final mud log (masterlog) & geological report	EOWR, subsurface team
Wireline Logging	Final log prints & digital data (LAS/DLIS)	EOWR, petrophysics, regulator
Completion & Well Test	Completion diagram & well test report	Well handover, production
Waste Management	Waste manifest & disposal certificates	Environmental reporting
Perforating / Logging Sources	Explosives usage & return; radioactive source return	Permit close-out
All Partners	Final field tickets & invoice, performance data	Financial close-out, SQM

Tip: write submission deadlines into each contract; late partner reports delay the EOWR and the AFE closure.

Service Quality Meeting (SQM)

Prepare data
KPIs, NPT caused, incidents, reports

Review performance
Plan vs actual per service line

Two-way feedback
Operator ↔ partner, no blame

Agree actions
Owner & due date for each gap

Vendor rating
Input for next tender & contract

Example scorecard: Drilling Fluids

KPI	Weight	Score (1-5)
HSE performance	25%	5
Service quality / NPT caused	25%	3
Equipment & materials	15%	4
Personnel competence	15%	4
Reporting timeliness	10%	3
Invoice accuracy	10%	4
Weighted score	**100%**	**3.9**

When: after rig release per partner; quarterly during a multi-well campaign.

End of Well Report

01

Well Summary

Objectives, final status, as-built schematic

02

Time Analysis

Actual vs plan, NPT & ILT by section

03

Cost Analysis

Final cost vs AFE by category

04

Geology & Evaluation

Formation tops, logs, cores, tests

05

Engineering Performance

Bits, BHA, mud, casing & cement

06

HSE & Lessons Learned

Incidents, root causes, recommendations

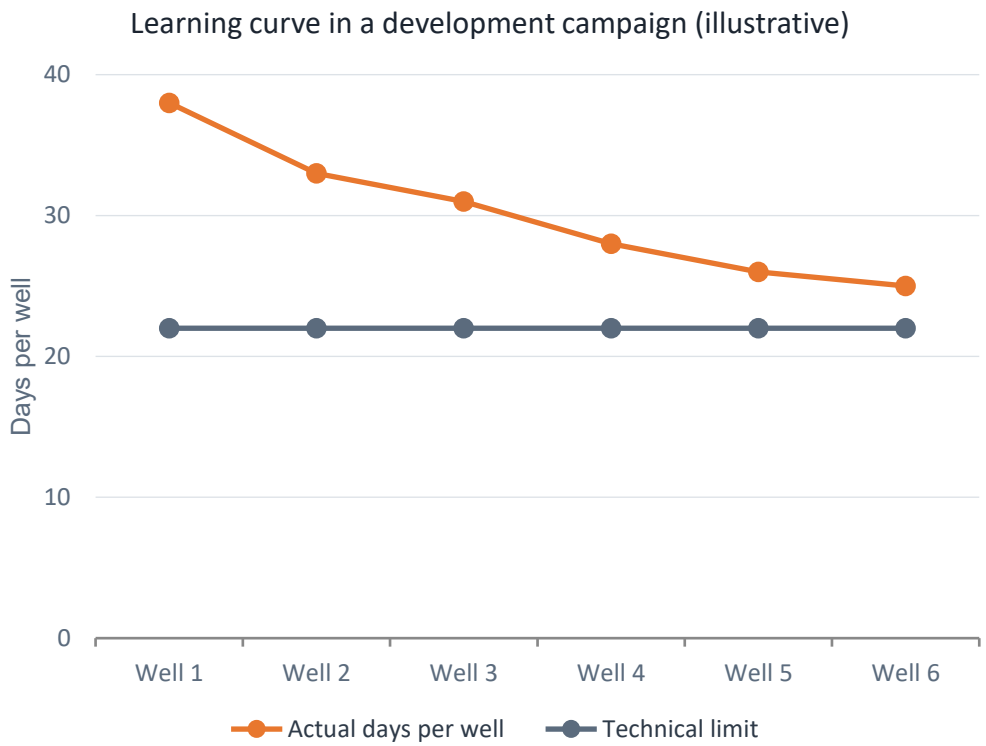
Purpose: the single permanent record of what was planned, what happened, and why. Typical target: issued within ~30 days of rig release (per company standard).

Final Performance Recap

KPI	Plan / Target	Actual	Variance	Status
Well duration	28 days (P50)	31 days	+3 days (+10.7%)	AMBER
Well cost	US$ 12.0 MM AFE	US$ 13.1 MM	+9.2% (within 10% cont.)	AMBER
NPT	≤ 8%	11.6%	+3.6 pts	RED
Total depth	3,500 m MD	3,510 m MD	Objective reached	GREEN
Safety	0 LTI	0 LTI, 14 near-miss reports	On target	GREEN
Data acquisition	Full log suite + 1 core	Achieved	On target	GREEN

Example well, consistent with the tracking charts. The recap feeds the EOWR and the post-mortem agenda.

The Success Multiplier



-34%

days per well from Well 1 to Well 6 when lessons are captured and applied

- Each well's lessons lower the next well's P50
- Gap to technical limit = remaining improvement potential
- Without capture, every well starts from zero

Knowledge Management Cycle



1. **Capture**
   AAR & post-mortem
2. **Analyze**
   Root cause, not blame
3. **Store**
   Lessons database
4. **Share**
   Peer review, SQM
5. **Apply**
   Next BOD & program

Lessons Learned Register (Example)

Event	Impact	Root Cause	Recommendation	Owner
Lost circulation in 12-¼" section	30 hrs NPT, US$ 0.5 MM	MW near frac gradient; LCM not pre-mixed	Pre-mix LCM pill; review PPFG margin in BOD	Drilling Eng.
Late delivery of 9-⅝" casing	Rig waiting 10 hrs	LLI tracking not updated weekly	Weekly LLI status in morning call	Logistics
Fast connections on night shift	ILT reduced 4 hrs/section	Crew pre-job planning & drills	Standardize as best practice	Company Man

Tip: record good practices too, not only failures. Every lesson needs an owner and a target document (BOD, program, procedure).

“A project is not finished when the bit stops turning. It is finished when the Lessons Learned are shared.”

- Adapted for DPM Excellence

Key Takeaways: The Full DPM Cycle

1 Initiation

Why drill?

Deliverable
Drilling Proposal

Gate
Gate 1 Approval

2 Planning

How to drill?

Deliverable
Basis of Design

Gate
Peer Review

3 Execution

Drill as planned, control deviations

Deliverable
Drilling Program

Gate
DWOP

4 Closing

What did we learn?

Deliverable
EOWR & Lessons

Gate
Handover & Close-out

Plan the whole cycle

from proposal to handover, not just to spud

Control, don't react

baseline, daily tracking, MOC

Close the loop

lessons learned make the next well better

PROJECT ASSIGNMENT

Project Assignment

Phase-1	Phase-2	Phase-3	Phase-4
Drilling Proposal	**Basis of Design**	**Drilling Program**	**N/A**
Gate	Gate	Gate	
Project Approval	**Peer Review**	**DWOP (Drilling Well on Paper)**	

This session: each group develops the Drilling Program and defends it at the DWOP. Phase-4 is covered in class only.

PMT Contribution

1. Subsurface Specialist

→ **Operations Geologist / Wellsite Geologist**

Drilling Program Deliverables:

- Drafts the Formation Tops & Coring/Logging Program.
- Establishes real-time correlation protocols for structural markers while drilling.

2. Drilling Design Engineer

→ **Rig Superintendent / Lead Drilling Engineer**

Drilling Program Deliverables:

- Writes the step-by-step Detailed Operational Sequence (from Spud-in to TD).
- Defines daily drilling parameters (RPM, Weight on Bit/WOB, flow rates).

PMT Contribution



3. Logistic Coordinator

→ Rig Move & Supply Chain Coordinator

Drilling Program Deliverables:

- Drafts the Rig Mobilization & Demobilization Schedule.
- Creates the Material Tracking Matrix for Long Lead Items (LLI) to ensure seamless delivery to the rig floor.



4. Risk & HSE Engineer

→ Rig HSE Officer / Safety Advisor

Drilling Program Deliverables:

- Develops the HSE Bridging Document between the Operator and Drilling Contractor.
- Prepares the Emergency Response Plan (ERP) and Rig Safety Induction protocols.

PMT Contribution

5. Project Contract & Economist

→ Cost Controller & Contract Specialist

Drilling Program Deliverables:

- Builds the Daily Cost Tracker Template and AFE curve monitoring spreadsheet.
- Defines invoice verification workflows for third-party service lines.

6. Project Manager (PM)

→ Drilling Operations Supervisor (Rig DSV / Company Man)

Drilling Program Deliverables:

- Establishes the Daily Drilling Report (DDR) Flowchart and communication hierarchy.
- Approves the final operational protocols and owns the Management of Change (MOC) trigger authority.

Data Requirements (The Inputs)

Final Engineering Designs (from Phase 2)

Approved Well Schematic
Finalized casing depths, hole sizes, and equipment specs.

Validated PPFG Data
The ultimate "Drilling Window" (Pore Pressure vs. Fracture Gradient).

Trajectory & Anti-Collision Report
The detailed directional plan and proximity analysis to offset wells.

Operational & Resource Logistics

Rig Technical Specifications
Rated capacities for pumps, hoisting, and the BOP stack configuration.

Service Company Contracts
Details of the specific equipment and fluids provided by the awarded vendors.

Material Inventory
On-site availability of casing joints, bulk cement, and mud chemicals.

Regulatory & HSE Framework

Approved Spud Permit
Validated legal permission to begin operations.

HSE Bridging Document
Aligned safety procedures between the Operator and the Drilling Contractor.

Minimal Content of a Drilling Program (The Outputs)

A General Information & Objectives

- Well Summary: names, coordinates, and elevations (KB/GL).
- Mission Statement: clear technical goals (e.g., drill to 3,500 m TD, perform wireline logs, and run a 7" liner).

B Detailed Operational Sequence

- The core of the document: a granular, phase-by-phase instruction list.
- Rig up & pressure testing → Spud-in & surface hole → Casing & cementing → BOP nipple-up → Formation evaluation (logging/coring).

Minimal Content of a Drilling Program (The Outputs)

C — Casing & Cementing Program

- Casing Tally: connection types, weights, and grades.
- Cementing Procedures: slurry volumes, densities, and pumping rates for zonal isolation.

D — Mud & Hydraulics Program

- Mud Properties: mud weight, viscosity, and chemical maintenance per section.
- Hydraulics Optimization: flow rates and nozzle selection for hole cleaning.

E — Well Control & Pressure Testing

- Kick Tolerance: gas influx limit the rig can safely circulate out.
- Barrier Verification: scheduled BOP tests and FIT/LOT.

Minimal Content of a Drilling Program (The Outputs)

F Monitoring & Project Controls

- Performance Tracking: Days vs. Depth (P50 curve) against which actual progress is measured.
- Reporting Protocols: DDR schedule and communication hierarchy.

G Management of Change (MOC)

- The formal process for field personnel to request deviations from the program due to unexpected subsurface findings.

DWOP Presentation Format: Walk-through

2'	3'	9'	2'	2'	2'
Opening	**Pre-spud**	**Section Walk-through**	**TD & Evaluation**	**Project Controls**	**Closing**
PM	Logistic + HSE	Surface → Interm. → Prod.	Subsurface	Contract & Economist	PM: Go / No-Go

In every hole section, four roles speak in sequence

Subsurface

Formation tops, hazards, pore & frac pressure

Drilling Engineer

Sequence, parameters, casing & cement, mud, well control

HSE

Critical PTW, SIMOPS, drills for this section

Logistic

Materials & services ready (look-ahead)

20' presentation + 10' panel Q&A. Show the links: one hazard should appear in the program, the look-ahead and the budget.

DWOP Assessment Rubric

Criteria	Weight	What the panel looks for
Technical soundness	**25%**	Safe, executable sequence consistent with BOD, PPFG and well design
Role mastery	**20%**	Each member explains own part and defends it under questioning
Integration	**20%**	Hazards traced across program, look-ahead, HSE and budget
Risk & contingency	**15%**	Top risks, triggers, Plan B and MOC path are clear
Project controls	**10%**	P50 curve, AFE curve, DDR flow and reporting defined
Delivery	**10%**	On time, clear flow, readable visuals

Sample panel challenges

- Subsurface: overpressure 200 m shallower than predicted. What changes?
- Drilling Eng.: what is the kick tolerance in the 8-½" section?
- Logistic: 9-⅝" casing arrives 5 days late. Impact on schedule?
- PM: who may approve a casing point change at 02:00?

Phase-3 Drilling Program

Diskusi Kelompok

Diskusi Kelompok

Berkumpul sesuai kelompok	Pastikan setiap peran PMT terisi
Bagi deliverable	Tiap anggota memegang bagian Drilling Program sesuai perannya
Draft Drilling Program	Susun kerangka A–G dan operational sequence bersama mentor
Presentasi draft	Paparkan alur program dan rencana persiapan DWOP

Back up

Rundown Sesi (3 Jam)

Menit	Durasi	Aktivitas	Detail Konten
0 - 10	10'	Opening & Review	Safety moment, agenda, review Peer Review BOD
10 - 55	45'	Phase-3: Execution & Monitoring	Execution core, performance tracking, NPT, AFE, information flow, MOC, safety
55 - 65	10'	Q&A	Tanya jawab Phase-3
65 - 75	10'	Break	Istirahat sejenak
75 - 105	30'	Phase-4: Closing	Handover, financial close-out, EOWR, performance recap, lessons learned
105 - 120	15'	Project Assignment	Brief Drilling Program, kontribusi PMT, input & output
120 - 165	45'	Diskusi Kelompok	Menyusun draft Drilling Program didampingi Mentor
165 - 180	15'	Closing & Mandate	Instruksi dokumen final & jadwal DWOP